THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which combination in a vegetarian diet gives good-quality protein? 1

(a) water with salt **(b)** sugar with butter
(c) cereals with pulses **(d)** only fruits

2. Why does a steel almirah become slightly magnetic over many years? 1

(a) it rusts **(b)** it gets heated
(c) it sees the sun **(d)** Earth's field aligns the tiny magnets

3. Which one of these questions leads to a hypothesis most directly? 1

(a) Why do wet clothes dry faster on a windy day?
(b) What is the date today?
(c) Who is taller?
(d) What do you want to eat?

4. Which of these shows interdependence between plants and animals? 1

(a) plants eat animals **(b)** animals use oxygen produced by plants
(c) animals make plants **(d)** plants hunt

5. Which type of food would best help recovery after a long illness? 1

(a) only sweets **(b)** spicy fried food
(c) easily digestible, nutritious food **(d)** no food

6. If the Earth's geographic north pole region behaves like a magnetic south pole, a compass's north needle is attracted to it because: 1

(a) unlike poles attract **(b)** wind
(c) gravity **(d)** like poles attract

7. Which of these is the best example of a testable question? 1

(a) Which sport is exciting?
(b) Is a mango tastier than a pear?
(c) Does temperature affect how fast sugar dissolves?
(d) Is blue a better colour than red?

8. Which group of features suits an animal living in the polar region? 1

(a) gills, fins, scales **(b)** thick fur, fat layer, white colour
(c) thin skin, long legs, hump **(d)** feathers, hollow bones, beak

9. Which pair correctly matches the deficiency and disease? 1

(a) iodine, anaemia **(b)** iron, scurvy
(c) vitamin A, night blindness **(d)** vitamin C, rickets

10. A compass kept on a table shows north in a wrong direction. The best explanation is: 1

(a) a magnet or iron object nearby **(b)** sunlight
(c) the table is wooden **(d)** noise

11. Raman used a simple setup to study light and received a Nobel Prize. This shows that: 1

(a) science is only for the rich
(b) Nobel prizes are easy
(c) careful observation matters more than costly equipment alone
(d) experiments are not needed

12. The roots of a banyan hanging from branches are called: 1

(a) tuberous roots **(b)** tap roots **(c)** prop roots **(d)** fibrous roots

13. Which food pair shows foods rich in fat and protein respectively? 1

(a) butter and dal **(b)** salt and sugar **(c)** rice and butter **(d)** apple and orange

14. A pin stuck to a magnet can also pick up another pin because: 1

(a) pins have glue **(b)** air presses them
(c) the pin acts as a temporary magnet **(d)** the pin is sticky

15. In a graph of plant height against days, a steady upward line shows: 1

(a) the soil is dry **(b)** the plant grows with time
(c) the plant is not changing **(d)** the plant is shrinking

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are growing children given more protein-rich food? 2

17. Why do we say 'magnets act through non-magnetic materials'? 2

18. Why do scientists use a control group? Give an example. 2

19. A student says 'all animals that fly are birds'. Is it correct? Give an example. 2

20. Why do children living near the poles with long dark winters need extra care for vitamin D? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why a mixture of cereal and pulse such as dal-rice gives a better meal than rice alone. 3

22. Explain how the magnetic compass helped in exploring new lands. 3

23. Compare the use of a thermometer with feeling with the hand to judge temperature. 3

24. Explain how the hump of a camel benefits it in the desert. 3

OR

Fruits and vegetables lose vitamins when cut and stored for long. Why and what to do? 3

25. How can you show the Earth's magnetism with a simple experiment? 3

OR

Describe two ways in which a scientific idea may be proved wrong. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Three students measured the time of 20 pendulum swings: 28 s, 30 s, 32 s. Find the average, the time for one swing, and say why 20 swings are timed. 5

OR

Describe how the structure of a bird's body (including beak, feet, bones, feathers) is adapted for its way of life with examples. 5

27. Plan a balanced day's menu for a 12-year-old, justify each meal's nutrients and mention what to avoid. 5

OR

A box has three unlabelled bars: a magnet, an iron bar and a copper bar. Describe a method with only the bars to identify each. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND